

Ten Lectures on Integral Lattices

Lecture Notes

1 Lattices

Definition 1. *A lattice is a free $\mathbb{Z}$ -module L of finite rank with a symmetric bilinear form $b: L \times L \rightarrow \mathbb{Q}$. It is integral if $b(L, L) \subseteq \mathbb{Z}$.*

2 Gram matrices

For a basis $e_1, \dots, e_n$ of L the Gram matrix is $G = (b(e_i, e_j))_{i,j}$, and

$$\det L = \det G.$$

A change of basis by $P \in \mathrm{GL}_n(\mathbb{Z})$ replaces G by $P^T G P$.

3 Dual lattices

The dual lattice is $L^\vee = \{x \in L \otimes \mathbb{Q} : b(x, L) \subseteq \mathbb{Z}\}$. For integral L we have $L \subseteq L^\vee$ and $[L^\vee : L] = |\det L|$.

4 Discriminant forms

For an even lattice the discriminant group $A_L = L^\vee/L$ carries the quadratic form

$$q_L(x + L) = \tfrac{1}{2}b(x, x) \bmod \mathbb{Z}.$$

5 Root lattices

The root lattice A_n is $\{x \in \mathbb{Z}^{n+1} : \sum_i x_i = 0\}$, with $\det A_n = n + 1$. The lattice D_n is $\{x \in \mathbb{Z}^n : \sum_i x_i \equiv 0 \pmod{2}\}$.

6 The lattice E_8

Theorem 1. $E_8 = D_8 \cup (D_8 + \frac{1}{2}(1, \dots, 1))$ is the unique even unimodular positive definite lattice of rank 8, and it has 240 roots.

7 Theta series

The theta series of a positive definite lattice is $\theta_L(q) = \sum_{x \in L} q^{b(x,x)/2}$. For E_8 ,

$$\theta_{E_8}(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n.$$

8 Genus

Two lattices are in the same genus if $L \otimes \mathbb{Z}_p \cong M \otimes \mathbb{Z}_p$ for every prime p and $L \otimes \mathbb{R} \cong M \otimes \mathbb{R}$.

9 The mass formula

The Smith–Minkowski–Siegel mass formula states

$$\sum_{M \in \text{gen}(L)} \frac{1}{|\mathcal{O}(M)|} = \text{mass}(L).$$

10 Niemeier lattices

There are exactly 24 even unimodular positive definite lattices of rank 24; the Leech lattice is the only one without roots.